Minh Le

Calgary, AB | +1 (825) 437-1645 | benny.minh@gmail.com | minhl6.github.io/portfolio | linkedin.com/in/minh-le-26b0103b1

Second-year mechanical engineering student interested in sustainable technology, who enjoys bridging theory with real, practical builds. Looking for opportunities to develop renewable energy hardware such as solar and electric.

Education

University of British Columbia, Vancouver **Expected graduation: May 2029**
BASc, Mechanical Engineering · Second Year · Cumulative GPA: 4.00/4.33 (A) · Dean's List

Projects

Two-Speed Hand-Crank Generator **June 2026 – August 2026**
Personal project

- Designed and built a **custom-made** hand-crank generator that sets its output voltage through gear selection instead of regulator electronics, **charging a phone** from hand power alone.
- Built a two-speed gearbox with a constant-mesh dog clutch that shifts under load; proved with logged data that each gear selects a distinct voltage (high gear ~1.9× low), modeled in **SolidWorks** and **3D printed** in PLA with **hand-machined** shafts.
- Characterized the **hand-wound axial-flux generator** with an **Arduino**, **INA219**, and **Hall effect sensor**; measured drivetrain efficiency up to ~**48%** around 42-52 crank RPM.

Leader-Follower Robotic Arm **May 2026 – June 2026**
Personal project

- Designed and built a 4-DOF arm that mirrors a hand-held twin in real time, with record and playback. Both arms were modelled in **SolidWorks** and **3D printed** in PLA.
- Designed a custom **Arduino shield PCB** in **KiCad**, manufactured by **JLCPCB**.
- Wrote three-mode firmware with sensor smoothing; fixed voltage dips and swapped to metal-gear servos for load.

Rainwater Harvester: System Simulation & Design in Excel **March 2026 – April 2026**
School project · 5-person team

- Built the topography model and a **5-year** daily water-balance **simulation** tracking system reliability.
- Modeled on-demand flow and pump-to-storage computationally from pipe friction, elevation, and pump curves.
- Placed **2nd of 10** in studio at **63.66% satisfaction** on the coordinators' withheld-weather simulation.

Technical Skills

Design/CAD: SolidWorks, Fusion 360, PrusaSlicer (slicing software), technical drawing (orthographic, isometric)
Electronics: PCB design in KiCad (schematic, board layout, trace sizing), Arduino, soldering, basic circuits & sensors
Programming: Embedded C/C++ (Arduino), Python, MATLAB, VS Code
Fabrication: 3D printing (FDM), manual fabrication, sheet-metal fabrication
Productivity/Data: Excel-based system modeling & data analysis, Microsoft Office, Google Workspace

Job Experience

Volleydome **May–Aug 2026, Jan 2024 – Jun 2025**
Club Volleyball Coach *Calgary, AB*

- Planned and led practices to build athlete skills, teamwork, and confidence.
- Gave individual feedback and communicated with athletes, parents, and staff.

Volleydome **Mar 2023 – Dec 2023**
Volunteer *Calgary, AB*

- Supported event setup, managed logistics, and helped participants with questions throughout the day.

Additional

Languages: English · Availability: Full Availability · Driver's licence: Class 5 (Alberta)